

Week 9 Progress Report & Project Documentation

Project Title: CARISURG Triage Guardian - Clinical AI Risk Identification System for Urgent Triage

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1. Project Overview

CARISURG Triage Guardian is an Artificial Intelligence (AI)-assisted Clinical Decision Support System (CDSS) designed to support emergency department (ED) triage. The system analyses patient demographic information, presenting complaints and physiological observations collected during triage to generate evidence-based recommendations that assist nurses in prioritising patients according to clinical urgency.

Unlike autonomous AI systems, CARISURG Triage Guardian follows a Human-Computer Interaction (HCI) approach, ensuring that healthcare professionals remain responsible for all final clinical decisions. The AI functions as a decision-support tool by highlighting high-risk patients, explaining the factors influencing its recommendation and providing clinicians with timely information to support safe and efficient triage.

2. Project Background

Previous phases of the project focused on understanding the emergency department dataset, evaluating data quality and identifying suitable predictive models.

Key milestones completed include:

- Exploratory data analysis of the Mercer emergency department dataset.
- Missingness analysis and data-quality assessment.
- Feature selection and clinical interpretation.
- Development of Logistic Regression and Decision Tree baseline models.
- Model optimisation using Random Forest.
- Comparison of predictive performance using accuracy, precision, recall and F1-score.
- Selection of the preferred model based on clinically appropriate evaluation metrics.

The evaluation prioritised recall for the highest-acuity patients, recognising that failing to identify critically ill individuals presents a greater patient safety risk than generating additional false-positive alerts.

3. Project Objectives

The project aims to:

- Support emergency department triage through AI-assisted decision support.
- Improve early identification of critically ill patients.
- Reduce the likelihood of under-triage.
- Provide explainable AI recommendations that clinicians can understand and evaluate.
- Integrate seamlessly into existing emergency department workflows.
- Maintain clinician oversight throughout the decision-making process.
- Improve patient safety while preserving efficient triage processes.

4. Current Development Progress

4.1 System Planning

The overall system architecture has been defined, including:

- project scope;
- intended users;
- workflow integration;
- functional requirements;
- AI integration points;
- expected system inputs and outputs.

The proposed implementation has been designed specifically for emergency department triage rather than as a general hospital information system.

4.2 Machine Learning Development

Machine learning development has progressed through multiple evaluation stages.

Completed activities include:

- Data preprocessing.
- Dataset exploration.
- Feature engineering.
- Logistic Regression baseline.
- Decision Tree baseline.
- Random Forest optimisation.
- Performance benchmarking.
- Clinical interpretation of evaluation metrics.

Model evaluation demonstrated that prioritising recall for critically ill patients provides a safer approach for emergency department triage because it reduces the likelihood of missing patients requiring immediate intervention.

4.3 User Interface Development

Development has now progressed into the implementation planning phase through the design of a clinician-facing interface.

The dashboard has been redesigned specifically for CARISURG AI and currently includes:

- application branding;
- navigation menu;
- emergency department overview;
- AI system status;
- quick actions;
- clinician notifications;
- dashboard summary cards.

The interface is intended to support triage nurses working in busy emergency department environments.

5. Proposed System Workflow

The proposed workflow is illustrated below.

1. Patient arrives at the emergency department.
2. Triage nurse performs the initial clinical assessment.

3. Patient demographics, vital signs and presenting complaint are entered into CARISURG AI.
4. The AI analyses the available information.
5. The system generates:
 - recommended triage category;
 - confidence score;
 - explanation of the recommendation;
 - suggested clinical action.
6. The triage nurse reviews the recommendation.
7. The recommendation is either accepted or overridden.
8. The final decision is recorded within the audit log.

This workflow ensures that responsibility for patient care remains with qualified healthcare professionals.

6. System Features

The proposed system currently includes the following functional components.

Dashboard

Provides an overview of emergency department activity, including patient statistics, AI system status and quick navigation tools.

AI Decision Support

Generates evidence-based recommendations using the selected machine learning model.

Explainable AI

Displays the clinical factors contributing to the AI recommendation, improving transparency and clinician trust.

Human Oversight

Allows clinicians to review, accept or override AI recommendations before any clinical action is taken.

Audit Trail

Records AI predictions, clinician decisions, overrides and timestamps to support governance and future quality improvement.

7. Human-Computer Interaction Considerations

CARISURG Triage Guardian has been designed as an HCI system because the primary user is the triage nurse.

The interface has been developed using several human-centred design principles, including:

- simple navigation;
- minimal cognitive load;
- clear visual hierarchy;
- colour-coded urgency indicators;
- high readability;
- rapid access to patient information;
- support for clinician judgement rather than automation.

These principles are intended to improve usability during high-pressure emergency department workflows.

8. Current System Modules

Module	Status
Authentication	Planned
Dashboard	Completed
New Patient Assessment	In Progress
AI Recommendation	Planned
Live Triage Queue	Planned

Module	Status
Critical Alerts	Planned
Patient History	Planned
Reports & Analytics	Planned
Audit Trail	Planned
Settings	Planned

9. Challenges Encountered

Several challenges have been encountered during the current development phase.

Initially, Figma was evaluated as the preferred interface design platform. However, editing community templates proved challenging because many components were grouped or locked. Canva was therefore adopted to facilitate rapid customisation while maintaining a professional healthcare appearance.

Another challenge involved transitioning from a predictive machine learning model to a complete clinical decision-support system. Considerable effort was therefore dedicated to ensuring that the proposed implementation reflects real emergency department workflows and supports clinicians rather than replacing them.

10. Remaining Work

The following tasks remain before project completion.

- Complete the New Patient Assessment interface.
- Design the AI Recommendation screen.
- Design the Live Triage Queue.
- Complete the Patient History module.
- Develop the Critical Alerts interface.
- Design the Reports and Analytics dashboard.
- Complete the HCI co-design canvas.
- Finalise the system requirements specification.
- Record the implementation walkthrough video.

- Prepare final documentation and presentation materials.

11. Conclusion

CARISURG AI has progressed from exploratory data analysis and predictive model development into the design of a human-centred clinical decision-support system. The project now combines machine learning, explainable AI and Human–Computer Interaction principles to support safer emergency department triage while preserving clinician oversight.

The current prototype establishes the foundation for a deployable emergency department application that integrates predictive analytics into existing clinical workflows. Future development will focus on completing the remaining interface modules, refining the user experience and preparing the system for demonstration as a fully integrated AI-assisted triage platform.